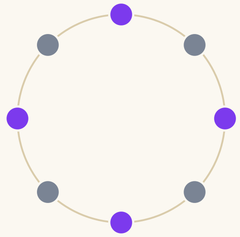
A CHAPTER IN PICTURES · II

The Compass and the Ear

Islamic geometric ornament, the circles of al-Khalīl, and the one mathematics beneath both

the compass



the ear

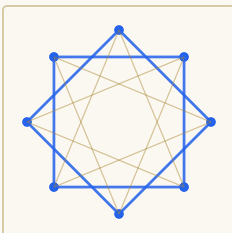
"A mathematician, like a painter or a poet, is a maker of patterns. If his patterns are more permanent than theirs, it is because they are made with ideas."

— G. H. HARDY, *A Mathematician's Apology* (1940)

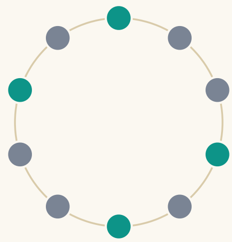
§1 Two Workshops, One Question

Hold two scenes side by side. In eighth-century Basra a lexicographer listens to verse after verse, hunting the law beneath the music, and finds that all of Arabic poetry hangs from five circles of repeating syllabic units. Some centuries later, in a workshop in Isfahan or Fez or Granada, a craftsman sets a compass on a blank panel, divides a circle into ten equal parts, and begins the strapwork that will cover a wall with stars.

Neither knows the other. One works in sound, the other in stone and glazed clay. Yet both have staked their art on the same wager: *that a small kit of parts, repeated under strict law, can carry unbounded beauty* — and both are asking, in their materials, the same question. What may repeat? Around what? And what does the law of repetition permit and forbid?



the star {8/2}, framed for a wall



the necklace of 10 — المختلف units

Fig. 1 — *The same object twice. Left, the eight-pointed star {8/2} as ornament — the seed of a thousand Islamic panels. Right, the necklace of al-Khalīl's first circle, from the previous chapter. Both are a circle, divided, and read under a rule.*

These are, historically, the two great abstract arts of Islamic civilization. Its poetry made almost no pictures of the world — it made *metres*, and gloried in them; its ornament, in the religious sphere, largely declined the image as well — and made *pattern*, and gloried in it. Two arts of pure structure, prized by one culture above nearly everything. This chapter is about what they share beneath the surface: not influence, not symbolism, but a common mathematics — discovered twice, with two different instruments. The craftsman's instrument was the compass. Al-Khalīl's was the ear.

§1 · CONTINUED

Begin with the kits, because their smallness is the point. The prosodic tradition builds every metre from four bricks — the cords and pegs of the previous chapter. And when modern researchers reconstructed how the most intricate five-fold Islamic patterns could have been laid out, they found that a kit of just five tiles — each edge the same length, each decorated with the same angled straps — suffices for masterpieces from Baghdad to Bukhara.

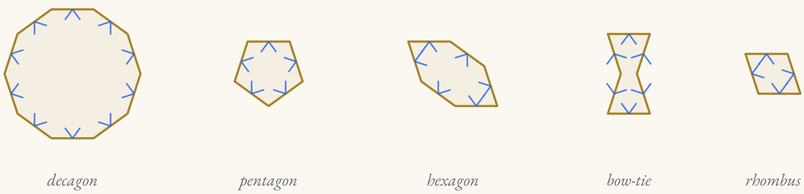


Fig. 2 — *The girih kit: decagon, pentagon, elongated hexagon, bow-tie, rhombus — five equal-edged tiles that generate the grand five-fold patterns. Below, the prosodist's kit from chapter I: four units that generate all sixteen metres.*

`0/`	`0//`	`/0/`	`0///`

A skeptic will say: any art has elements; bricks do not make a bond. True. The bond is what each tradition does *next*, and it is the same thing. Neither treats its kit as a box of free parts. Both submit the parts to a generating discipline — a circle, a division, a rule of assembly — so that the finished work is not an arrangement someone happened to like but a *consequence*: one member of a lawful family that could, in principle, be enumerated entire. Art by classification. The rest of this chapter is a walk through that shared discipline, from its shared vocabulary of stars to its two different laws of what symmetry is permitted — and to the one place each art touches something the other cannot reach.

§2 The Circle Beneath

Watch the craftsman's hands. Nearly every Islamic geometric panel, however bewildering its finished web, begins the same way: a circle; the circle divided into n equal arcs; the division points joined by a counting rule — every second point, every third — and then, crucially, the scaffolding *erased*. The construction circle never appears in the finished wall. What the viewer receives is the star; what generated the star is withdrawn.

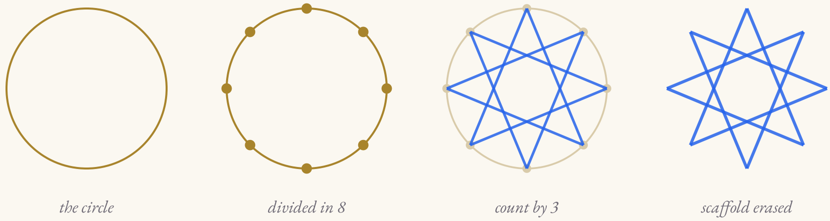


Fig. 3 — *The artisan's liturgy, in four acts: circle; division into eight; the count-by-three chords {8/3}; scaffold erased, star delivered. Every panel hides this circle the way a poem hides its metre-wheel.*

Now recall, from the previous chapter, how a metre is built: a circle of units; a starting point chosen on it; the units read off around the loop — and then the circle *disappears*. No poem ever exhibits its dāʾira. The audience receives a line of sound; the wheel that generated the line is a specialist's secret, drawn only in the prosodists' manuals.

The parallel is exact, and it runs deeper than method — it is a shared *attitude about where structure lives*. In both arts the generating object is a circle marked with a discrete rhythm of points; in both, the public artifact is a projection of that circle — onto a wall, onto a breath of time — and in both, the tradition kept the circles anyway, in a private literature of diagrams, because the makers understood that the erased scaffold, not the visible surface, is where the art's truth is stored.

§2 · CONTINUED

the wall, with its secret restored

the line, with its wheel restored

Fig. 4 — *Two erased scaffolds. Left: the finished star with its construction circle ghosted back in. Right: a line of* الطويل *with its necklace ghosted behind it — the dā'ira no listener ever sees.*

> ### The geometer among the artisans
>
> The bridge between mathematics and the workshop is not a modern conjecture; it is documented. In tenth-century Baghdad the geometer **Abū al-Wafā' al-Būzjānī** wrote *On the Geometric Constructions Necessary for the Artisan* — compass-and-straightedge recipes addressed directly to craftsmen — and describes meetings where geometers and artisans worked problems together. Five centuries later the **Topkapı scroll** (studied by Gülru Necipoğlu, 1995) preserves a working pattern-book: thirty metres of construction drawings, the scaffolding of walls that were built to hide it. The prosodists' manuals, with their drawn circles of metres, are the same genre of document for the art of time: the scroll of the erased wheels.

Keep this pairing — public surface, private circle — because it sharpens what "abstraction" means in both arts. Neither hides its structure out of secrecy; the structure is simply *prior* to any one appearance of it. A wall is one printing of a pattern; a poem is one printing of a metre. The circle is the plate from which the printings are pulled — and the two traditions, independently, both discovered that the plate deserved its own literature.

§3 A Shared Alphabet of Stars

If the two arts share a written character, it is the star polygon {n/k}: n points on a circle, every k-th point joined. The previous chapter met these stars as the *drawn signature of a stabilizer* — join each unit of a necklace to the unit its hidden symmetry carries it to, and the symmetry sketches {n/p}. Islamic ornament, working in space with a free choice of n and k, made the same figures its most beloved vocabulary:

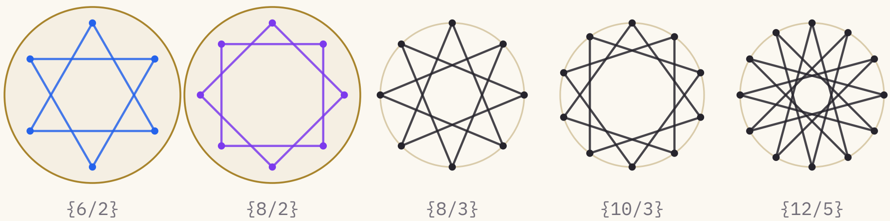


Fig. 5 — *Five citizens of the ornamental alphabet. The hexagram {6/2} and the eight-star {8/2} — marked in gold — are precisely the stabilizer stars of the circles of* المؤتلف *and* المتفق *from chapter I. The others, {8/3}, {10/3}, {12/5}, are the plane's luxuries: stars poetry's necklaces never happened to need.*

The construction is identical on both sides — the counting rule on a divided circle — and so is its deeper reading. A star {n/k} is not a picture *of* anything; it is the visible record of an action, the orbit of one point under "advance by k." When gcd(n, k) is 1 the star is one unbroken path; when it is d > 1, the star resolves into d interlocking polygons — the hexagram's two triangles, the eight-star's two squares, the {9/3}'s three triangles that chapter I drew for the circle of المجتلب. The interlace so characteristic of Islamic design — separate closed paths woven through one another — is, at the skeleton, the *coset structure* of a subgroup, worn as decoration.

§3 · CONTINUED

One coincidence deserves a page of its own.

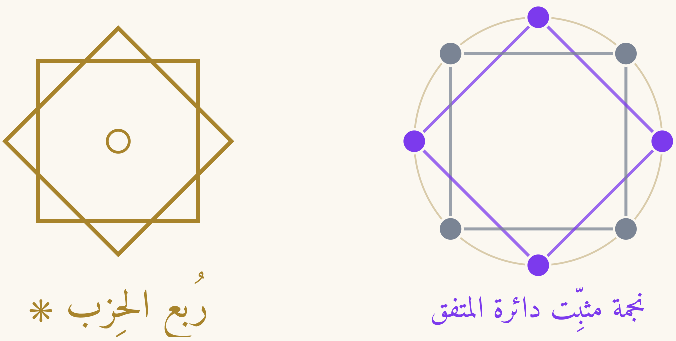


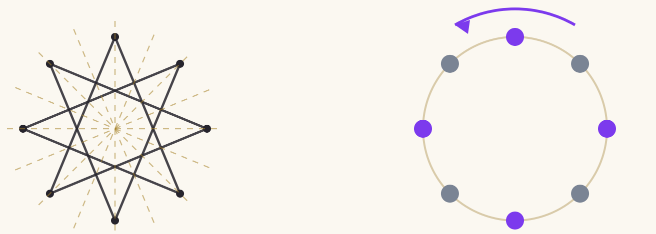
Fig. 6 — *Left: the rubʿ al-ḥizb ۞ — two interlocking squares, {8/2} — the mark that has divided the text of the Qur'an for a millennium. Right: the stabilizer star of the circle of* المتفق*, drawn by its own C_4 symmetry on its own eight units, exactly as in chapter I.*

They are the same figure. The ornamental mark that punctuates the divisions of scripture is, stroke for stroke, the star that the necklace of المتقارب and المتدارك draws when its rotational symmetry is made visible. No transmission connects them; nobody copied anything from anybody. Two practices — one dividing a sacred text into eighths for recitation, one dividing a circle of sounds into readings — arrived at the identical object because both were performing the identical mathematics: a cycle of eight, acted on by a count of two.

The same is true one circle earlier. The hexagram — *khātam*, the Seal, among the oldest and commonest devices in Islamic (and older) ornament — is {6/2}: two interlocked triangles. Chapter I drew it as the stabilizer of the circle of المؤتلف, where its two triangles came out *meaning* something: one triangle threads the three pegs, the other the three fāṣilas. The ornament and the metre agree on the figure; the metre, additionally, colours it. This is what a shared mathematics looks like when two arts wear it: not resemblance — *identity*, arrived at from opposite directions.

§4 Where the Mirrors Live

Here the two arts part company — and the parting is as instructive as the kinship. Take the craftsman's eight-fold rosette. Its symmetries include the eight rotations, but also eight *reflections*: mirror lines through the points and through the notches. Its full symmetry group is the dihedral D_8, sixteen moves strong. Ornament lives in space, and space does not care which way you walk past a wall.



D_8 — eight turns, eight mirrors

C_8 — eight turns, no mirror: time's arrow

Fig. 7 — *Left: the rosette's mirrors — eight axes, drawn as hairlines; with its eight rotations they make D_8. Right: the necklace's poverty — rotations only, C_8, because a rhythm is heard in one direction. Time is the wall you cannot walk back past.*

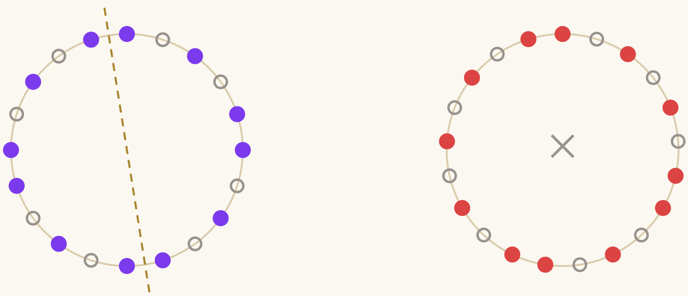
Now try to give the metre its mirror. Reflecting a rhythm means performing it backwards — and a verse recited backwards is not a verse; the ear receives sound strictly in time's order, and time has an arrow. So of the full dihedral wealth available to the eye, the ear can use exactly half: the rotations. That is why every group in the previous chapter was C_n and never D_n — not a modelling choice, but physics. The plane reflects; the moment does not.

Islamic art knows the difference from its own side. Most of its patterns embrace mirrors — but a strain of them refuses: pinwheel motifs in brickwork that spin without reflecting, whirling rosettes whose group is a bare C_n. And when the tradition does reflect *language*, in the mirror-calligraphy called *muthannā*, it reflects the written image of the words, never their sound: even art's boldest mirror stops at the ear.

§4 · CONTINUED

Which raises a question nobody in either tradition could have asked: *do the necklaces of al-Khalīl have mirrors they cannot hear?* Lay a metre's circle of struck-and-quiet letters on a wall, as ornament, so that reflection becomes lawful — is the reflected necklace the same necklace?

The answer, computed letter by letter across all five circles, is a small astonishment:



المتفق — achiral: the mirror is one of its own turns

المشتبه — chiral: no axis; its mirror is a stranger

Fig. 8 — *Two necklaces as pure letter-rings (● struck, ○ quiet). Left: the circle of* المتفق *— the hairline axis reflects the ring onto itself exactly. Four of the five circles carry such an axis. Right: the circle of* المشتبه *— no axis exists; its reflection is a necklace that occurs nowhere in the system. Chiral, like a hand.*

Four of the five necklaces are *achiral*: seen as ornament, each equals its own mirror image (after a turn). Their full symmetry in space is dihedral — richer than anything the ear, confined to rotations, can ever exercise. Poetry owns mirrors it cannot hear; only geometry can see them.

And the fifth circle — the exception — is المشتبه. Again. The circle that chapter I found stripped of rotational symmetry (C_1, the one-peg catastrophe) turns out to lack the mirror as well: it is the only *chiral* necklace in the system, asymmetric in every geometry that will have it. The tradition named it "the resembling, the confusable"; the mathematics keeps agreeing, from every direction one approaches, that it is the strangest object in the five — the necklace that is like nothing, not even its own reflection.

§5 The Art of the Strip

Between the rosette (a point) and the wallpaper (a plane) lies the border: pattern running along a strip. Mathematics counts exactly **seven** ways a strip pattern can be symmetric — the seven frieze groups — and Islamic ornament, in its Qur'anic borders, its cornice bands, its woven edgings, uses them all.

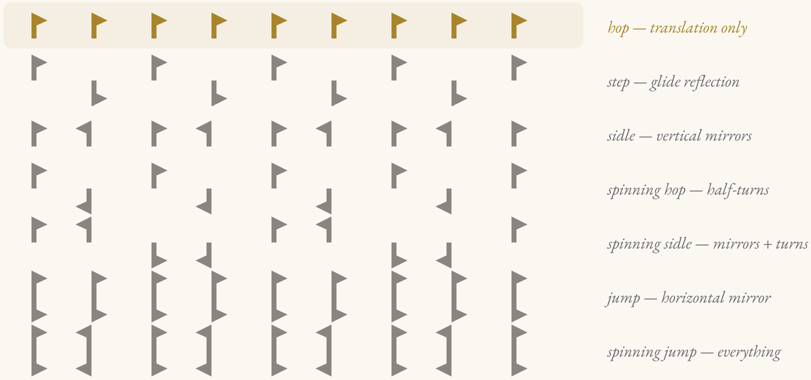


Fig. 9 — *The seven friezes, one motif. Translation alone; glide; vertical mirrors; half-turns; and their combinations. A verse of poetry is also a strip — but sound admits only the first row (gold): translation, foot after foot. Every other row needs a mirror or a turn that time refuses.*

A line of metred verse is a frieze in sound: the foot repeats, فعولن مفاعيلن فعولن مفاعيلن, a translation rolling left along time. But of the seven strip symmetries, six require some flip — a vertical mirror (the strip read backwards), a horizontal one (the strip turned upside-down), a half-turn, a glide. For rhythm, "backwards" is unperformable and "upside-down" does not even parse. The ear's border-art owns one row of the seven-row catalogue.

Musicians, it is true, have played with the forbidden rows — the crab canon runs a melody against its own retrograde. But notice where that game is actually played: on the *page*, in notation, where time has been laid out as space and can be read in either direction. It is muthannā for sound — a mirror applied to time's picture, because it cannot be applied to time.

§5 · CONTINUED

Climb one dimension more and the plane opens. Patterns that repeat in two directions — wallpapers, pavements, the tiled dado of every madrasa — admit exactly **seventeen** symmetry types: the wallpaper groups, proved complete by Fedorov in 1891 and again by Pólya in 1924. Islamic practice, centuries earlier, had walked deep into that catalogue by pure craft — the Alhambra alone exhibits, depending on the census and its rules of evidence, between eleven and all seventeen. The count is argued; the astonishment is not: a complete abstract classification, explored to its corners by artisans with no notion that a classification existed.

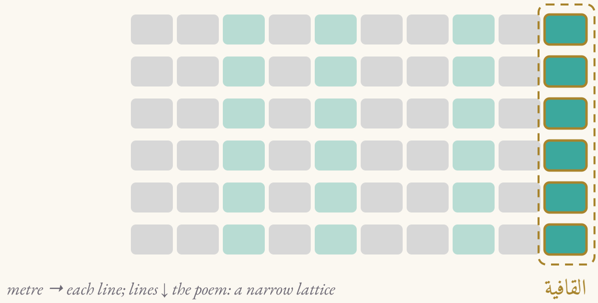


Fig. 10 — *Poetry's timid second dimension. A qaṣīda is a stack of metrically identical lines — horizontal translation by the foot, vertical translation by the verse — with one column pinned: the qāfiya, the rhyme every line must end on (gold). A wallpaper, but the poorest one on the shelf; sound's plane is narrow.*

Poetry does gesture at the plane. A classical qaṣīda repeats its metre line under line — a vertical translation — and nails the right-hand margin with a single rhyme, sounded at every line's end for a hundred lines. As pattern, that is a lattice: the foot repeating one way, the verse the other. But it is a lattice with almost no further symmetry available — no mirrors, no turns, for the reasons of §4 — while the eye's plane blossoms into seventeen kinds of richness. Where the strip gave sound one row of seven, the plane widens the gap to one of seventeen. The eye's freedom is two-dimensional; the ear walks a line.

§6 Each Geometry Legislates

Chapter I found the necklace's law of permitted symmetry: a stabilizer's size must divide the bead-count — Lagrange's ladder of divisors, nothing in between. The plane has its own law, and it is stranger. A pattern that repeats in two directions can carry rotations of order 1, 2, 3, 4 and 6 — *and no other*. No five-fold centre, no seven-fold, ever, in any periodic pattern: the crystallographic restriction.

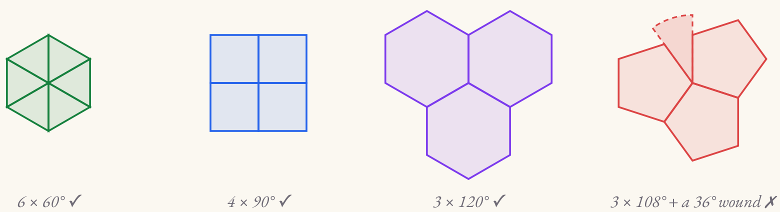


Fig. 11 — *Why five is outlawed. Triangles, squares and hexagons close neatly around a point; regular pentagons leave a 36° wound that no fifth pentagon can heal. What cannot tile a neighbourhood cannot anchor a lattice.*

WORLD	ITS OBJECTS	ITS LAW OF SYMMETRY
the necklace (time's loop)	metres	stabilizer order divides n — any n welcome
the plane (space's field)	tilings	rotation orders 1, 2, 3, 4, 6 only

Fig. 12 — *Two legislatures. Each geometry writes its own statute of lawful symmetry; neither statute is negotiable from inside.*

Set the two statutes side by side and a fine irony appears. The necklace is permissive about n — al-Khalīl's circles run 6, 8, 9, 10 beads with no objection — but strict about what fits inside n (divisors only). The plane is permissive about size and strict about kind: whatever the pattern, five-fold hearts are contraband. Each art worked happily for centuries inside its own statute. But one of them, eventually, decided to smuggle.

§6 · CONTINUED

Islamic art loved the five-fold star — the very order the plane's law forbids in a repeating field. The first response was diplomatic: build ten-point rosettes as local medallions, embedded in a field whose actual repetition obeyed the law. But in the girih work of Iran a bolder strategy matured, and at the Darb-i Imam shrine in Isfahan (1453) it reached a configuration that stopped modern physicists cold.

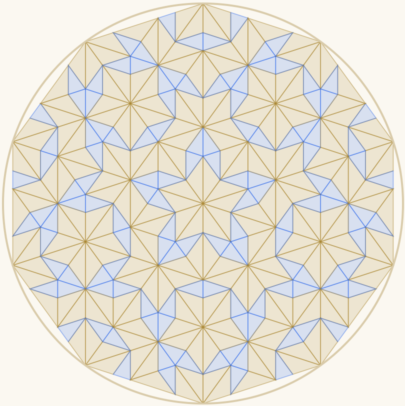


Fig. 13 — *Order without repetition: a five-fold patch built by golden-ratio subdivision (the Penrose rhombi, drawn here by their triangle-halves). Perfectly lawful, never periodic — the loophole in the plane's statute. The Darb-i Imam girih realizes a pattern of this family.*

In 2007 Peter Lu and Paul Steinhardt showed that the Darb-i Imam pattern, generated by subdividing large girih tiles into small ones, maps onto a nearly perfect *quasicrystalline* tiling — the kind of five-fold, never-repeating order that Roger Penrose constructed in the 1970s and that earned a Nobel prize when found in matter itself. Honesty requires the caveats: predecessors (Emil Makovicky had argued a Penrose-like reading of the 1197 Maragha tower years earlier), critics (Peter Cromwell and others favour more traditional construction readings), and the qualifier "nearly." What no one disputes is the mathematics of the object: a pattern with perfect long-range law and no repeating cell — *structure beyond periodicity*. The plane's statute forbade five-fold repetition; the craftsmen found the sense in which order never needed repetition at all.

§7 Ground and Figure

Stand in any great Islamic interior and notice how the symmetry is *deployed*. The dado runs a strict periodic field — calm, endless, self-effacing. And against that ground, the design breaks its own symmetry exactly where it wants your soul to look: the miḥrāb, the medallion, the inscription band where geometry yields to the moving line of the word. Perfect repetition is the *ground*; the *figure* is a lawful violation of it.

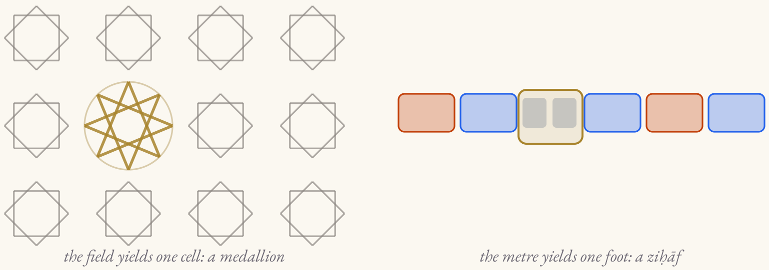


Fig. 14 — *The grammar of emphasis, in both arts. Left: a periodic star-field with one cell broken open into a medallion — the eye goes where the symmetry breaks. Right: a metric line with one foot varied by ziḥāf (gold) — the ear pricks where the pattern bends.*

Poetry's prosody has precisely this grammar. The metre is the ground — fixed, contractual, the listener's horizon of expectation. And the ziḥāf system (the licensed variations that the confusion-map of this book's companion software anatomizes) is the figure: a foot lightened here, a syllable quieted there — lawful violations that keep the line alive against its own regularity. The classical critics valued a verse neither slack nor rigid; the classical builders tiled a wall neither chaotic nor dead. Both arts discovered that beauty does not reside in symmetry itself. It resides at the *edge* of symmetry, where a deep order lets itself be locally disturbed.

Chapter I stated the same economics from the theory's side: the inverse law, by which the most symmetric circle is the poorest in metres. Symmetry is ground because symmetry is cheap sameness; distinction — a metre, a medallion, a line that a listener will remember — is bought by breaking it.

§8 A Rosetta Stone

It is time to lay the two arts side by side and read the correspondence entire — concept for concept, not mood for mood:

ORNAMENT (SPACE)	METRE (TIME)
motif — the repeated device	the foot (تفعيلة)
construction circle, divided in n	the dāʾira of n units
unit cell / fundamental region	the period p — chapter I's arc
rosette symmetry D_n (with mirrors)	stabilizer C_p (mirrors unhearable)
star polygon {n/k}, interlaced cosets	stabilizer star {n/p} on the necklace
frieze — 7 lawful strips	the verse line — 1 lawful strip
wallpaper — 17 lawful planes	the qaṣīda's narrow lattice
crystallographic restriction (2,3,4,6)	Lagrange's divisor ladder
catalogue explored by craft (Alhambra)	catalogue derived by ear (21 rotations)
girih: order beyond periodicity	— (time's line is too narrow)
erased scaffold; the Topkapı scroll	silent circle; the prosodists' diagrams
ground/figure: field and medallion	metre and ziḥāf

Fig. 15 — *The correspondence. Where a row is blank, the blank is itself information: each art reaches one place the other cannot.*

Read the last column of blanks closely, because the table is honest about asymmetry. Ornament has no muhmal cells — nothing in a wall corresponds to a metre that is lawful but unused, because walls do not have a canon of sixteen. And metre has no quasicrystal — a one-dimensional ear cannot be paid in the plane's strange coin. The two arts overlap on a great shared core and then each keeps a private wonder. That, and not any mystical union, is what "one mathematics, two instruments" means.

§9 Two Instruments, One Discipline

A compass is a machine for making equalities: every point it reaches is the same distance from its foot. An ear is a machine for detecting them: two durations, two stresses, two cadences — the same, or not. One instrument constructs symmetry; the other audits it. Everything in this chapter follows from handing the same discipline — *pattern under law* — to those two instruments and letting each work its native medium.

The compass, in space, could build its groups outright: all seventeen wallpapers waiting to be walked into, mirrors free of charge, and, at the tradition's far edge, the quasi-periodic loophole. The ear, in time, had the narrower kingdom — cyclic groups only, one frieze row, a thin lattice — but it did something in that kingdom the compass never needed to do: it *derived the complete catalogue*. Twenty-one rotations; sixteen adopted; five vacancies named or noted. The artisans explored their catalogue the way travellers explore a country. Al-Khalīl *proved his map*.

And both, it should be said plainly, were practicing mathematics by the standards of their own civilization's philosophers. When Ibn al-Haytham analysed beauty in the *Book of Optics*, he located it not in things but in relations — proportion, composition, the harmony of parts — exactly the register in which a tiling or a metre is beautiful. The culture possessed a theory of why lawful pattern pleases; its two great abstract arts are that theory, executed.

What this argument is not

No secret geometry of the poets, no numerology of the tilers, no code. Craftsmen thought in recipes and repertoire; prosodists thought in sounds and licenses. The claim is structural, and therefore checkable: examined later, with group theory in hand, both repertoires turn out to be organized by the same objects — actions, orbits, stabilizers, restrictions — because both crafts obeyed their media, and the media obey mathematics. The convergence is the discovery.

§10 The Formalism Arrived Late — Twice

The previous chapter ended on a date: the 1830s, when Galois gave the name *group* to what an eighth-century ear had already used. This chapter must end on two more. In 1891 Fedorov proved there are exactly seventeen wallpaper groups — and the walls of the Alhambra had been standing, their catalogue substantially explored, for five and a half centuries. In 1974 Penrose published his aperiodic five-fold tilings — and the girih spandrels of Darb-i Imam had been holding their nearly-quasicrystalline order since 1453.

Three times, then, in one civilization's orbit, the same silhouette: a practice runs ahead of its mathematics by centuries, guided by nothing but a disciplined instrument and an appetite for lawful beauty. The instrument differs — ear, compass, compass again — the mathematics recovered from the practice is one subject. It is hard to imagine better evidence for the old suspicion that mathematics is less invented than *met*: wherever a craft binds itself honestly to a medium, the structure of that medium surfaces in the work, whether or not anyone yet has words for it.

Symmetry is not a style.
It is what any patient art finds at the bottom of its material.

And so the two arts of this chapter, which never consulted each other, are colleagues after the fact. The prosodist turning his circle of sounds and the muhandis turning his compass were holding the two ends of one object — the group — across the distance between time and space. Neither could see the other end. Both carved their end so faithfully that, twelve centuries later, the join is seamless.

For the Hands and the Ears

Draw one

Take a real compass. Draw a circle; keep the radius; walk it round the circumference — it fits exactly six times (this is the compass's own theorem). You now have {6/2} and {6/3} for free: join alternate points for the Seal, all diameters for the wheel. For the eight-star ⊛, bisect twice instead. You have repeated, by hand, the first page of every pattern scroll — and the construction al-Khalīl's diagrams perform in sound.

Hear one

In the Interactive Arud Explorer (**arudi.midadalfikr.com**), open the Mathematics view and choose دائرة المتفق: watch its C_4 symmetry draw the two squares of Fig. 6 on the necklace itself. Then open the Rhythm Clock and press play on the same circle: what you now hear — the four-fold return of the drum — is the ⊛ as a sound. One object; your eye took it from the wall, your ear from the wheel.

Break one, lawfully

In the Confusion Map view, play الكامل and apply الإضمار while it sounds: a single stroke falls silent — ground and figure, ziḥāf as the medallion of time. Then find the same gesture on any great wall: the one cell where the field yields to the miḥrāb.

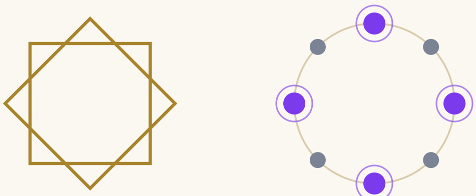


Fig. 16 — *One object, two organs: the eight-star as compass-work and as drum-work — the necklace of المتفق with its four-fold pulse marked.*

Glossary

girih گره
“knot”: the Persian strapwork style; also the five-tile kit that generates its five-fold patterns (§1, §6).

zellij / zillīj زليج
cut-tile mosaic of the Maghrib and al-Andalus.

rosette
a point-symmetric star device; symmetry C_n or D_n (§4).

dihedral group D_n
the n rotations *and* n reflections of an n-fold figure — twice the cyclic group C_n of chapter I (§4).

chiral / achiral
distinct from its mirror image / equal to it. Four necklaces are achiral; المشتبه alone is chiral (§4).

frieze groups
the seven symmetry types of strip patterns; sound can perform exactly one (§5).

wallpaper groups
the seventeen symmetry types of doubly periodic plane patterns (Fedorov 1891, Pólya 1924) (§5).

crystallographic restriction
periodic plane patterns admit rotation orders 1, 2, 3, 4, 6 only — the plane’s counterpart of Lagrange’s ladder (§6).

quasiperiodic
perfectly ordered, never repeating; Penrose tilings, and (nearly) the Darb-i Imam girih (§6).

khātam خاتم
the Seal: hexagram {6/2} — the stabilizer star of (3§) المؤتلف.

rubʿ al-ḥizb ✳
Qur’anic division mark: two squares, {8/2} — the stabilizer star of (3§) المتفق.

muthannā مثنّى
mirror calligraphy: reflecting the image of words, since their sound cannot be reflected (§4).

muqarnas, miḥrāb, dado
vault ornament of tiered cells; prayer niche; the lower wall zone that carries the periodic field (§7).

qāfiya قافية
the monorhyme pinning the qaṣīda’s end-column (§5).

ziḥāf زحاف
licensed metrical variation — the figure against the metre’s ground (§7; anatomized in the companion’s confusion map).

Sources & Notes

On the geometry of Islamic ornament. Abū al-Wafāʾ al-Būzjānī, *On the Geometric Constructions Necessary for the Artisan* (10th c.); G. Necipoğlu, *The Topkapı Scroll: Geometry and Ornament in Islamic Architecture* (1995); J. Bonner, *Islamic Geometric Patterns* (2017); S. J. Abas & A. S. Salman, *Symmetries of Islamic Geometrical Patterns* (1995); B. Grünbaum & G. C. Shephard, *Tilings and Patterns* (1987).

On the Alhambra census. E. Müller's 1944 thesis counted eleven wallpaper groups; later studies argue for thirteen up to all seventeen. The dispute concerns rules of evidence (colour, interlace, restoration), not the mathematics.

On girih and quasiperiodicity. P. J. Lu & P. J. Steinhardt, "Decagonal and Quasi-crystalline Tilings in Medieval Islamic Architecture," *Science* 315 (2007); E. Makovicky on the Maragha tower, in *Fivefold Symmetry* (1992); P. R. Cromwell's critical re-examination, *Math. Intelligencer* (2009). The chapter reports the finding with its debates attached.

On pattern, beauty, and mathematics. G. H. Hardy, *A Mathematician's Apology* (1940); H. Weyl, *Symmetry* (1952); Ibn al-Haytham, *Kitāb al-Manāẓir*, Bk II, on beauty as a relation of parts. Fedorov (1891) and Pólya (1924) for the seventeen; the frieze seven are classical.

Colophon. Companion to "The Turn That Changes Nothing"; same types (EB Garamond, Cormorant Garamond, Amiri, IBM Plex Mono), same 6×9 leaf. All prosodic claims are computed against the sequence data of the Interactive Arud Explorer: the stabilizer stars {10/5}, {6/2}, {9/3}, {8/2}; and the chirality census of §4 — the letter-necklaces of المختلف، المؤتلف، المجتلب، المتفق each equal their own reversal up to rotation (mirror offsets 7, 2, 2, 2 respectively), while المشتبه alone does not. Star figures are drawn by the counting rule itself; the five-fold patch of Fig. 13 by golden-ratio triangle subdivision. Prosodic symbols keep classical order: `0//` reads strike–strike–rest, from the right.